

food insecurity

Quinn

2026-06-05

```
source("/Users/natalieli/Desktop/ECON484 Project/FoodInsecurity_Unemployment (1).R")
```

```
## [merge] analysis rows=1478 | matched a 2021 SNAP value=1470 (99.5%)
## [sample] N=1215 young adults | unemployment rate=0.184 | SNAP-2021 rate=0.174
##
## (1a) OLS: unemployed ~ foodstamp_2021 (bivariate)
##
## t test of coefficients:
##
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.152390   0.011886  12.821 < 2.2e-16 ***
## foodstamp_2021 0.179363   0.034713   5.167 2.779e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## (1b) OLS: + lagged controls [SE clustered by 1968 family]
##
## t test of coefficients:
##
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.0625591  0.2452960  8.4084 < 2.2e-16 ***
## foodstamp_2021  0.0799559  0.0335983  2.3798 0.0174786 *
## age_2021        -0.0021338  0.0044742 -0.4769 0.6335083
## female          0.0149000  0.0231520  0.6436 0.5199743
## educ_yrs_2021   -0.0292064  0.0062426 -4.6785 3.216e-06 ***
## emp_2021_working -0.1903621  0.0268340 -7.0941 2.216e-12 ***
## days_unemp_2021  0.0111648  0.0103766  1.0760 0.2821608
## flourish_2021   -0.0033341  0.0036030 -0.9254 0.3549608
## soc_anxiety_2021 -0.0382240  0.0107710 -3.5488 0.0004019 ***
## log_fam_expend_2021 -0.1102261  0.0198478 -5.5536 3.441e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R^2 = 0.166

## (2) LASSO (LPM, alpha=1)
## lambda.min=0.00758 lambda.1se=0.05345
## foodstamp_2021 coef @lambda.1se: 0.0000 (dropped)
## Features kept @lambda.1se:
## [1] "educ_yrs_2021"      "emp_2021_working"    "log_fam_expend_2021"
```

```
##
## forced-in foodstamp_2021 coef @lambda.1se: 0.1448

## DMRL results i.e. ATE of 2021 SNAP receipt on P(unemployed)
##      ATE      SE p_value
## 1 0.0654 0.0434 0.1317
##
## ATE = change in P(unemployed in 2023) from the 2021 family having received
## food stamps (D: 0 -> 1), flexibly controlling for the lagged characteristics.
```

Main Results

```
dml_result <- data.frame(
  Model = "Double ML",
  ATE = round(irm$coef, 4),
  SE = round(irm$se, 4),
  p_value = round(irm$pval, 4)
)

knitr::kable(dml_result, caption = "ATE of 2021 SNAP receipt on 2023 unemployment")
```

Table 1: ATE of 2021 SNAP receipt on 2023 unemployment

	Model	ATE	SE	p_value
foodstamp_2021	Double ML	0.0654	0.0434	0.1317

Descriptive Statistics

```
desc <- data.frame(
  N = nrow(d),
  Unemployment_Rate = mean(d$unemployed),
  SNAP_2021_Rate = mean(d$foodstamp_2021),
  Mean_Education_Years = mean(d$educ_yrs_2021),
  Mean_Age = mean(d$age_2021)
)

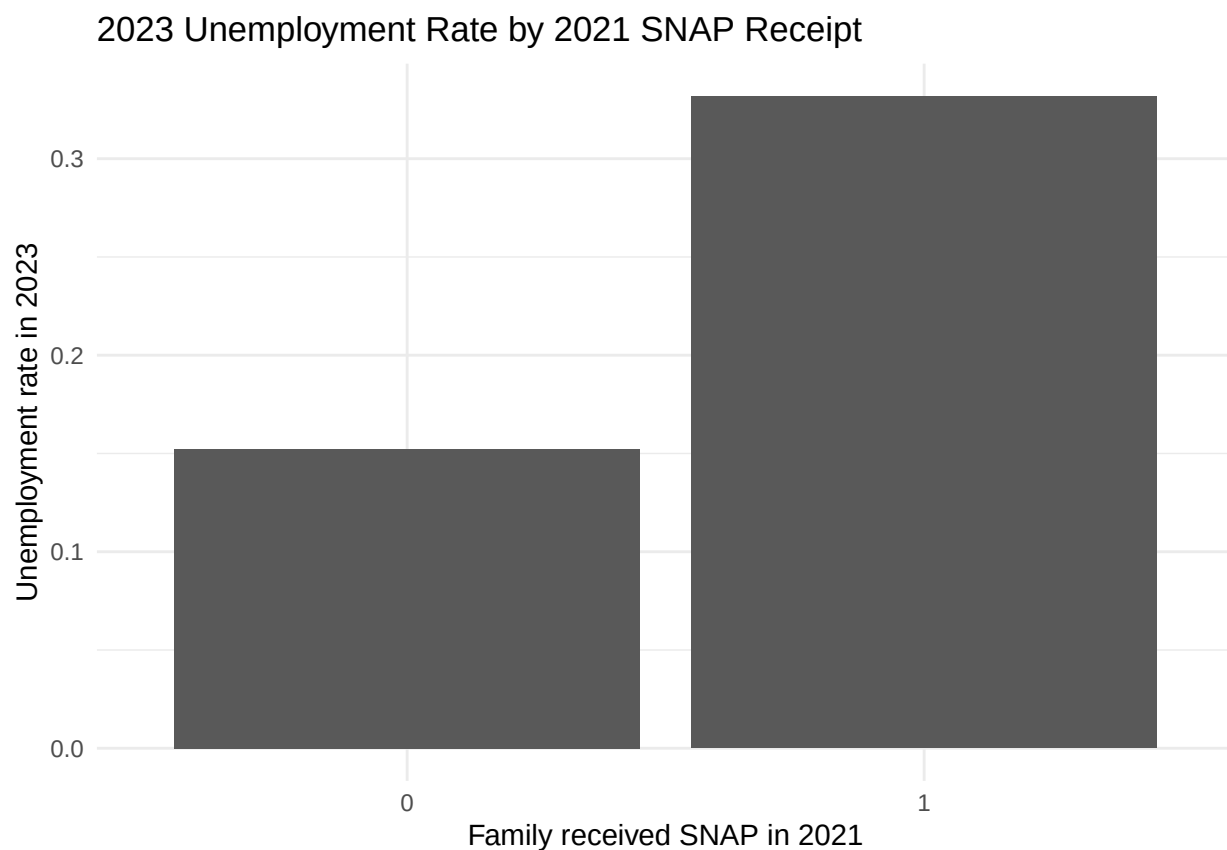
knitr::kable(desc, digits = 3)
```

N	Unemployment_Rate	SNAP_2021_Rate	Mean_Education_Years	Mean_Age
1215	0.184	0.174	13.058	22.194

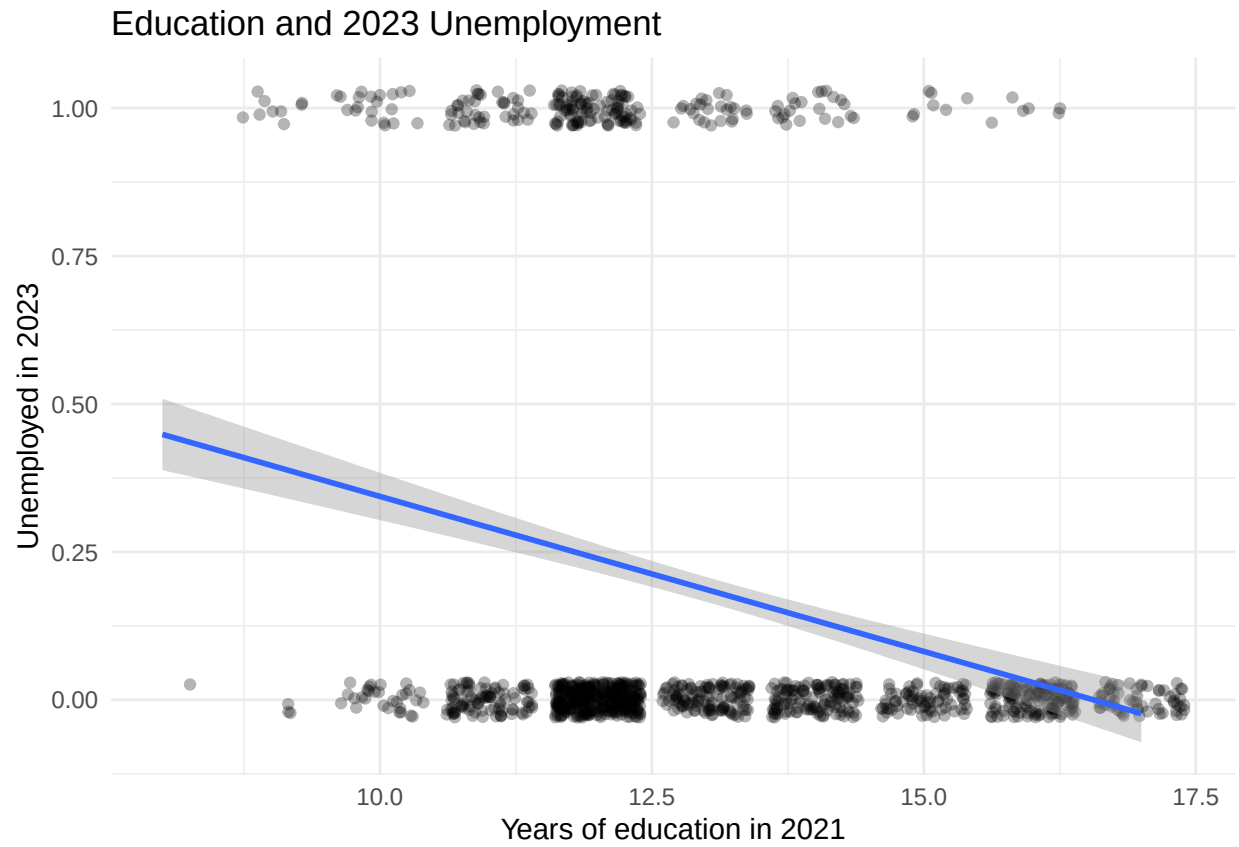
Graphs

```
library(ggplot2)

ggplot(d, aes(x = factor(foodstamp_2021), y = unemployed)) +
  stat_summary(fun = mean, geom = "bar") +
  labs(
    title = "2023 Unemployment Rate by 2021 SNAP Receipt",
    x = "Family received SNAP in 2021",
    y = "Unemployment rate in 2023"
  ) +
  theme_minimal()
```



```
ggplot(d, aes(x = educ_yrs_2021, y = unemployed)) +
  geom_jitter(height = 0.03, alpha = 0.3) +
  geom_smooth(method = "lm") +
  labs(
    title = "Education and 2023 Unemployment",
    x = "Years of education in 2021",
    y = "Unemployed in 2023"
  ) +
  theme_minimal()
```



““